

```
ubuntu@ip-172-31-13-169:~/devops-build$ sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:4 https://download.docker.com/linux/ubuntu noble InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
```

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
    https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
    https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
    /etc/apt/sources.list.d/jenkins.list > /dev/null"
sudo apt update
sudo apt install jenkins
```

```
ubuntu@ip-172-31-13-169:~/devops-build$ sudo systemctl status jenkins
```

```
● jenkins.service - Jenkins Continuous Integration Server
```

```
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
```

```
   Active: active (running) since Mon 2026-01-05 17:38:25 UTC; 43s ago
```

```
 Main PID: 62556 (java)
```

```
    Tasks: 42 (limit: 1121)
```

```
  Memory: 296.3M (peak: 313.4M)
```

```
    CPU: 14.131s
```

```
 CGroup: /system.slice/jenkins.service
```

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-094b4c0d3d5bc1b82	SSH ▼	TCP	22	Cus... ▼	Q 0.0.0.0/0 X	Delete
sgr-00d25a702bb2d430c	HTTP ▼	TCP	80	Cus... ▼	Q 0.0.0.0/0 X	Delete
—	Custom TCP ▼	TCP	8080	An... ▼	Q 0.0.0.0/0 0.0.0.0/0 X	Delete

Add rule



- [+ New Item](#)
- [📄 Build History](#)

[✎ Add description](#)

[Build Queue](#) [▼](#)

No builds in the queue.

[Build Executor Status](#) [0/2 ▼](#)

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

[Create a job](#) [+](#)

Set up a distributed build

[Set up an agent](#) [🖥️](#)

[Configure a cloud](#) [☁️](#)

[Learn more about distributed builds](#) [📖](#)

🔍 docker



Install



Install

Name ↓

Released

Health



Docker 1308.vff6e33248305

Cloud Providers

Cluster Management

docker

1 mo 15 days ago

100

This plugin integrates Jenkins with Docker

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

commons-lang3 v3.x Jenkins API

 Success

Icons API

 Success

Folders

 Success

OWASP Markup Formatter

 Success

ASM API

 Success

JSON Path API

 Success

Structs

 Success

Pipeline: Step API

 Success

commons-text API

 Success

Token Macro

 Success

Build Timeout

 Success

```
ubuntu@ip-172-31-13-169:~/devops-build$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword  
1f898005f4314ad8bcdff386be73eda9  
ubuntu@ip-172-31-13-169:~/devops-build$
```



```
ubuntu@ip-172-31-13-169:~$ cd devops-build
```

```
ubuntu@ip-172-31-13-169:~/devops-build$ sudo usermod -aG docker jenkins
```

```
ubuntu@ip-172-31-13-169:~/devops-build$ sudo systemctl restart jenkins
```

```
ubuntu@ip-172-31-13-169:~/devops-build$
```

New Credentials

Kind

Username with password



Scope



Global (Jenkins, nodes, items, all child items, etc)



Username

deepesh79

☐

Treat username as secret



Password

.....

ID



dockerhub-creds

+

New Item



New Item

Enter an item name

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Activate Windows

Go to Settings to activate Windows





GitHub project

Project url 

`https://github.com/deepeshchandnani/devops-build.git`

Advanced 



GitHub hook trigger for GITScm polling



Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM



SCM ?

Git



Repositories ?

Repository URL ?



https://github.com/deepeshchandnani/devops-build.git

Credentials ?

deepesh79/*****



+ Add

Branches to build ?

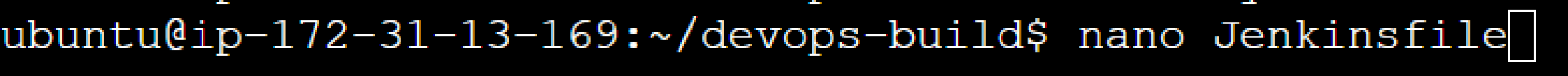
Branch Specifier (blank for 'any') ?



*/dev

Save

Apply



```
pipeline {  
    agent any  
  
    environment {  
        DOCKERHUB_CREDENTIALS = 'dockerhub-creds'  
        DEV_IMAGE = 'deepesh79/dev:latest'  
        PROD_IMAGE = 'deepesh79/prod:latest'  
    }  
  
    stages {  
  
        stage('Checkout') {  
            steps {  
                checkout scm  
            }  
        }  
    }  
}
```

```
stage('Build & Push DEV Image') {
    when {
        branch 'dev'
    }
    steps {
        script {
            docker.withRegistry('', DOCKERHUB_CREDENTIALS) {
                sh 'docker build -t $DEV_IMAGE .'
                sh 'docker push $DEV_IMAGE'
            }
        }
    }
}

stage('Build & Push PROD Image') {
    when {
```

```
stage('Build & Push PROD Image') {  
    when {  
        branch 'master'  
    }  
    steps {  
        script {  
            docker.withRegistry('', DOCKERHUB_CREDENTIALS) {  
                sh 'docker build -t $PROD_IMAGE .'  
                sh 'docker push $PROD_IMAGE'  
            }  
        }  
    }  
}
```

```
ubuntu@ip-172-31-13-169:~/devops-build$ git status
```

```
On branch dev
```

```
Changes to be committed:
```

```
  (use "git restore --staged <file>..." to unstage)
```

```
    new file:   Jenkinsfile
```

```
    modified:   build.sh
```

```
    modified:   deploy.sh
```

```
    modified:   dockerfile
```

```
ubuntu@ip-172-31-13-169:~/devops-build$ █
```

```
ubuntu@ip-172-31-13-169:~/devops-build$ git commit -m "all files"
[dev ec0fec3] all files
Committer: Ubuntu <ubuntu@ip-172-31-13-169.ap-south-1.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:
```

```
git config --global --edit
```

After doing this, you may fix the identity used for this commit with:

```
git commit --amend --reset-author
```

```
4 files changed, 50 insertions(+), 3 deletions(-)
create mode 100644 Jenkinsfile
mode change 100644 => 100755 build.sh
mode change 100644 => 100755 deploy.sh
ubuntu@ip-172-31-13-169:~/devops-build$
```



```
ubuntu@ip-172-31-13-169:~/devops-build$ git remote add origin https://deepeshchandnani:github_pat_11AUNT5BQ0v1lckhRcOe90_df1siNqc6X72imqZI3R
VKxzTtH11M2wwuK9beoEBRaf4ECWFXLXcIOeol3A@github.com/devops-build.git
ubuntu@ip-172-31-13-169:~/devops-build$ git push origin dev
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 887 bytes | 887.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/deepeshchandnani/devops-build.git
    572e4cc..ec0fec3  dev -> dev
ubuntu@ip-172-31-13-169:~/devops-build$
```

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

Webhooks / Add webhook

We'll send a `POST` request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, `x-www-form-urlencoded`, *etc*). More information can be found in [our developer documentation](#).

Payload URL *

Content type *





#1

9:25 AM



```
Started by user admin
Obtained Jenkinsfile from git https://github.com/deepeshchandrani/devops-build.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-build-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Check out SCM)
[Pipeline] checkout
selected git installation does not exist. Using default
The recommended git tool is: NONE
using credential dockerhub-creds
Cloning the remote git repository
Cloning repository https://github.com/deepeshchandrani/devops-build.git
  > git init /var/lib/jenkins/workspace/devops-build-pipeline # timeout=10
Fetching upstream changes from https://github.com/deepeshchandrani/devops-build.git
  > git --version # timeout=10
  > git --version # 'git version 3.13.0'
using GIT_CREDENTIAL to set credentials
  > git fetch --tags --force --progress -- https://github.com/deepeshchandrani/devops-build.git +refs/heads/*:refs/remotes/origin/* # timeout=10
  > git config remote.origin.url https://github.com/deepeshchandrani/devops-build.git # timeout=10
  > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
  > git rev-parse refs/remotes/origin/dev^{commit} # timeout=10
Checking out Revision e87a987ee87a278cd08a77be867a7d173ce8 (refs/remotes/origin/dev)
  > git config core.sparsecheckout # timeout=10
  > git checkout -f e87a987ee87a278cd08a77be867a7d173ce8 # timeout=10
Commit message: "test dev pipeline"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] checkout
selected git installation does not exist. Using default
The recommended git tool is: NONE
using credential dockerhub-creds
  > git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/devops-build-pipeline/.git # timeout=10
Fetching changes from the remote git repository
  > git config remote.origin.url https://github.com/deepeshchandrani/devops-build.git # timeout=10
Fetching upstream changes from https://github.com/deepeshchandrani/devops-build.git
  > git --version # timeout=10
  > git --version # 'git version 3.13.0'
using GIT_CREDENTIAL to set credentials
  > git fetch --tags --force --progress -- https://github.com/deepeshchandrani/devops-build.git +refs/heads/*:refs/remotes/origin/* # timeout=10
  > git rev-parse refs/remotes/origin/dev^{commit} # timeout=10
Checking out Revision e87a987ee87a278cd08a77be867a7d173ce8 (refs/remotes/origin/dev)
  > git config core.sparsecheckout # timeout=10
  > git checkout -f e87a987ee87a278cd08a77be867a7d173ce8 # timeout=10
Commit message: "test dev pipeline"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build & Push DEV Image)
stage "Build & Push DEV Image" skipped due to when conditional
[Pipeline] getTest
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build & Push PROD Image)
stage "Build & Push PROD Image" skipped due to when conditional
[Pipeline] getTest
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Stage "Build & Push PROD Image" skipped due to when conditional

```
[Pipeline] getContext
```

```
[Pipeline] }
```

```
[Pipeline] // stage
```

```
[Pipeline] }
```

```
[Pipeline] // withEnv
```

```
[Pipeline] }
```

```
[Pipeline] // withEnv
```

```
[Pipeline] }
```

```
[Pipeline] // node
```

```
[Pipeline] End of Pipeline
```

Finished: SUCCESS